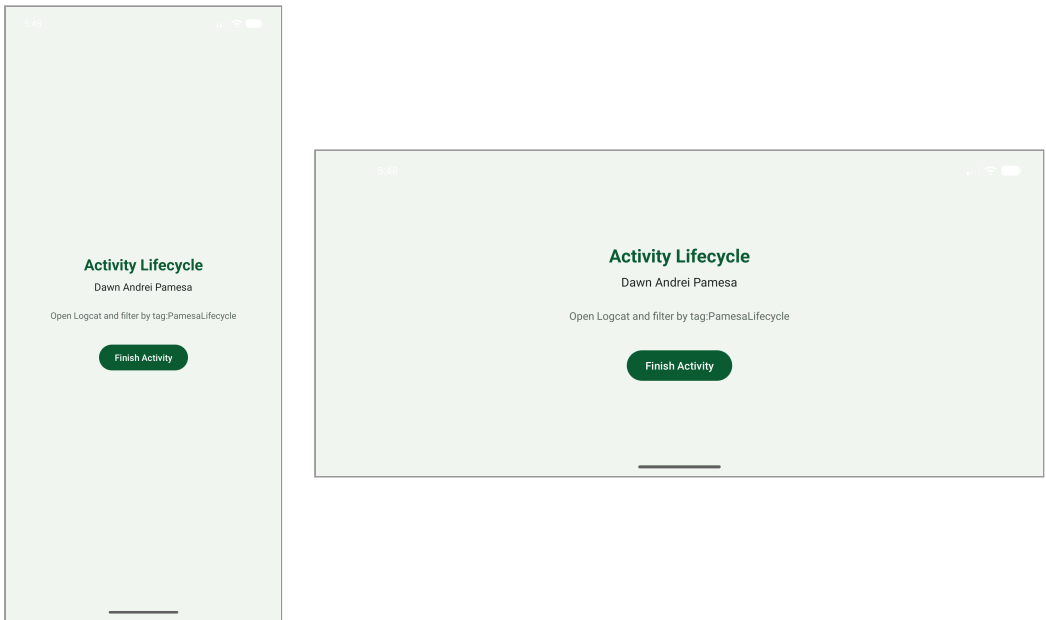


Technical 2: Android Activity Lifecycle

Lifecycle Observation Report

Application Overview

The app has one Activity, MainActivity, that overrides all seven lifecycle callbacks: onCreate(), onStart(), onResume(), onPause(), onStop(), onRestart() and onDestroy(). Each callback calls Log.d() with the tag PamesaLifecycle and the message Dawn Andrei Pamesa is calling <method>(), so all messages can be filtered in Logcat with tag:PamesaLifecycle. A Finish Activity button calls finish() so that onDestroy() can be triggered on purpose. The app was tested on the Pixel 10 Pro emulator.



Observation Table

#	Action Performed	Callback Sequence Observed	Explanation
1	Launch the app	onCreate() → onStart() → onResume()	The system creates the Activity, makes it visible, then brings it to the foreground so the user can interact with it.
2	Press Home	onPause() → onStop()	The Activity loses focus first (onPause), then is fully hidden by the home screen (onStop). It stays in memory, so it is not destroyed.

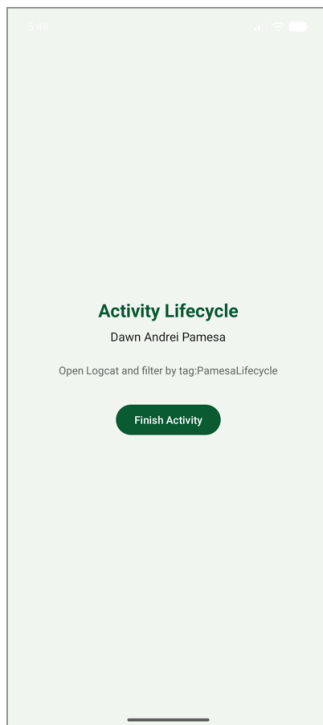
3	Reopen the app	onRestart() → onStart() → onResume()	The Activity was only stopped, so it comes back through onRestart() instead of onCreate(), then becomes visible and interactive again.
4	Switch to another app (Settings)	onPause() → onStop()	Settings covers the whole screen, so the Activity is paused and then stopped, the same as pressing Home.
5	Switch back to the app	onRestart() → onStart() → onResume()	The stopped Activity is brought back to the front, so it restarts, starts and resumes without being created again.
6	Rotate the device to landscape	onPause() → onStop() → onDestroy() → onCreate() → onStart() → onResume()	Rotation is a configuration change. By default Android destroys the Activity and creates a new one so the layout reloads for the new orientation.
7	Press Back	onPause() → onStop() → onDestroy()	Back tells the system the user is done with this screen, so the Activity is paused, stopped and destroyed, and the home screen shows.
8	Launch the app again	onCreate() → onStart() → onResume()	The previous Activity was destroyed, so a brand new one starts with onCreate() instead of onRestart().
9	Tap the Finish Activity button	onPause() → onStop() → onDestroy()	finish() ends the Activity on purpose, so onDestroy() is guaranteed to run here, unlike when the system kills the process.

Logcat Evidence

Each item shows the emulator screen right after the action and the Logcat messages it produced, filtered by the PamesaLifecycle tag. All seven callbacks appear: onRestart() in items 3 and 5, and onDestroy() in items 6, 7 and 9.

1. Launch the app

Emulator



Logcat (filter: tag:PamesaLifecycle)

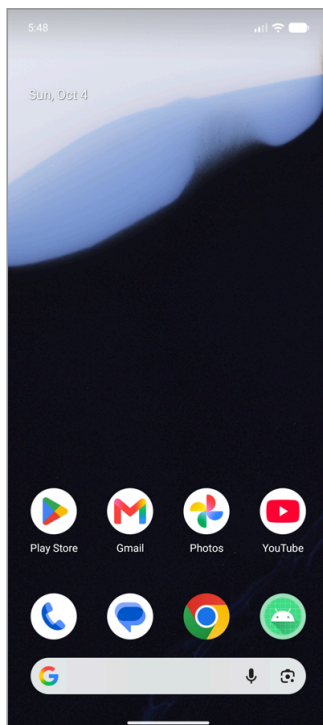
```
05:47:59.511 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onCreate()

05:47:59.543 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStart()

05:47:59.551 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onResume()
```

2. Press Home

Emulator



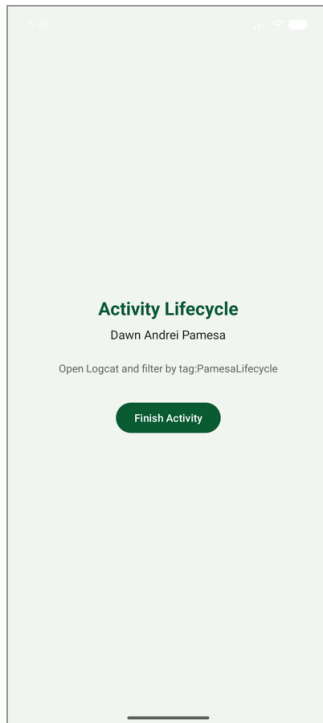
Logcat (filter: tag:PamesaLifecycle)

```
05:48:03.402 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onPause()

05:48:03.973 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStop()
```

3. Reopen the app

Emulator



Logcat (filter: tag:PamesaLifecycle)

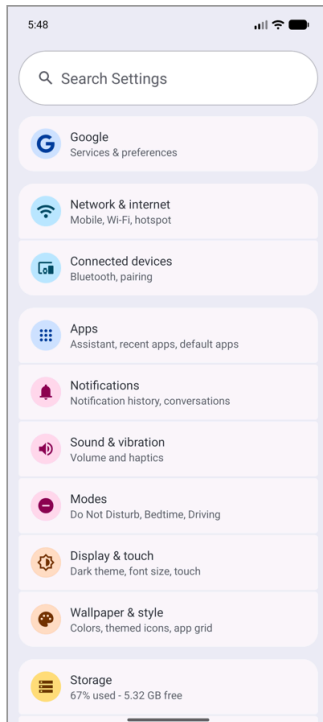
```
05:48:07.558  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStart()

05:48:07.559  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStart()

05:48:07.559  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onResume()
```

4. Switch to another app (Settings)

Emulator



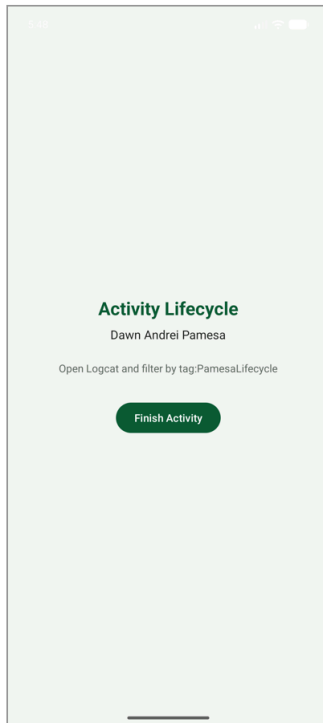
Logcat (filter: tag:PamesaLifecycle)

```
05:48:11.371  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onPause()

05:48:12.038  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStop()
```

5. Switch back to the app

Emulator



Logcat (filter: tag:PamesaLifecycle)

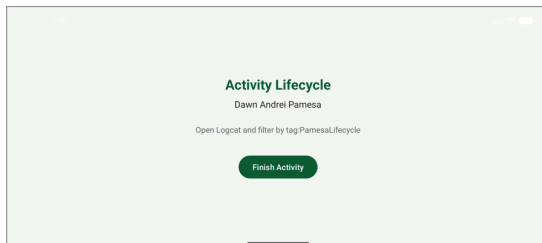
```
05:48:14.981  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStart()

05:48:14.982  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStart()

05:48:14.982  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onResume()
```

6. Rotate the device to landscape

Emulator



Logcat (filter: tag:PamesaLifecycle)

```
05:48:18.723  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onPause()

05:48:18.735  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStop()

05:48:18.736  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onDestroy()

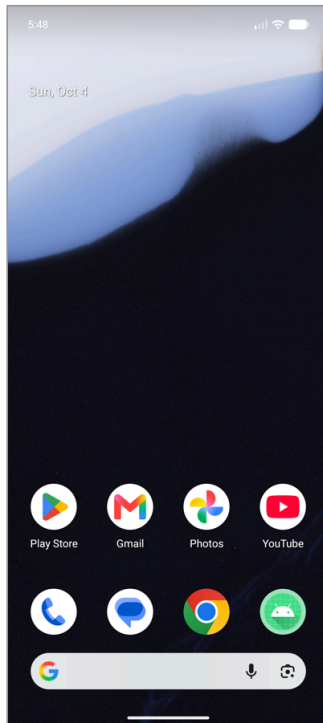
05:48:18.783  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onCreate()

05:48:18.785  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStart()

05:48:18.786  9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onResume()
```

7. Press Back

Emulator



Logcat (filter: tag:PamesaLifecycle)

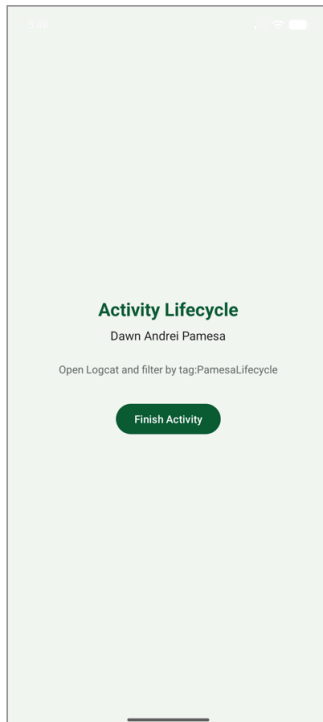
```
05:48:25.487 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onPause()

05:48:26.048 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStop()

05:48:26.051 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onDestroy()
```

8. Launch the app again

Emulator



Logcat (filter: tag:PamesaLifecycle)

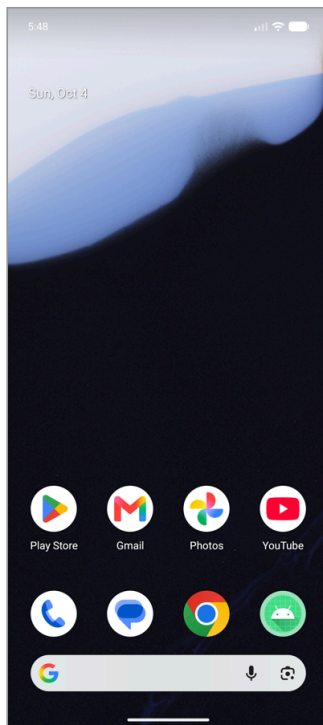
```
05:48:29.615 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onCreate()

05:48:29.616 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStart()

05:48:29.618 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onResume()
```

9. Tap the Finish Activity button

Emulator



Logcat (filter: tag:PamesaLifecycle)

```
05:48:35.993 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onPause()

05:48:36.586 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onStop()

05:48:36.589 9629-9629
D PamesaLifecycle
Dawn Andrei Pamesa is calling onDestroy()
```

Conclusion

The Activity lifecycle follows a fixed order. Opening an Activity always runs `onCreate()`, `onStart()` and `onResume()`, and leaving it always runs `onPause()` before `onStop()`. Whether the Activity comes back through `onRestart()` or `onCreate()` depends on whether it was only stopped or fully destroyed: Home and switching apps only stop it, while Back, `finish()` and rotation destroy it. Rotation was the most surprising result, because the same screen is destroyed and rebuilt, which means any data not saved would be lost. Since Android does not guarantee `onDestroy()` when it kills the process, important data should be saved in `onPause()` or `onStop()` instead of waiting for `onDestroy()`.